Interpretable Film Simulation Emulation from Paired Images Using Matrix, Tone Curve, and Residual 3D LUT

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***Abstract*—This paper presents a data-driven method for emulating camera film simulations from paired images without access to the proprietary in-camera pipeline. The proposed approach decomposes the transformation into three stages: a 3×3 color matrix, a luminance-based tone curve, and a compact residual 3D LUT. The matrix captures global color shifts, the tone curve models brightness and contrast changes, and the LUT corrects remaining nonlinear color differences. Experiments on paired Standard and Classic Chrome images show that the method achieves slightly better accuracy than a direct 33×33×33 LUT while using only a 17×17×17 residual LUT. The resulting framework is compact, interpretable, and suitable for practical photographic style reproduction.**

***Keywords—film* simulation, 3D LUT, color matrix, tone curve, paired image learning**

I. INTRODUCTION

Digital cameras often provide multiple built-in film simulation modes that alter color, contrast, and tone to create a distinctive visual style. Examples include Fujifilm simulations such as Classic Chrome, Velvia, and Provia. These profiles are widely used because they allow photographers to achieve a desired appearance directly in-camera without extensive post-processing. However, the internal algorithms used to generate these simulations are proprietary and are not publicly available.

As a result, reproducing a film simulation from an ordinary image remains a challenging problem. A common solution is to learn a direct mapping between a source image and a target style using a three-dimensional lookup table (3D LUT). A sufficiently large LUT can approximate complex color transformations and is widely used in image processing and color grading. Nevertheless, a direct LUT approach has two major limitations. First, a high-resolution LUT requires a large number of cells, increasing memory usage and making the transformation difficult to interpret. Second, the LUT acts as a black box because it does not explicitly separate global color shifts, tone changes, and local nonlinear corrections.

This paper proposes a data-driven method for film simulation emulation using paired source and target images. Instead of learning the entire transformation with a single large 3D LUT, the proposed approach decomposes the problem into three stages: a global 3×3 color matrix, a luminance-based tone curve, and a compact residual 3D LUT. The color matrix models the overall rotation of colors between the source and target profiles, while the tone curve captures differences in brightness and contrast. A smaller residual LUT is then used to correct the remaining local nonlinear color differences.

The proposed pipeline is trained directly from paired images captured using a Standard profile and the target film simulation profile. Experimental results show that the method achieves slightly higher reconstruction accuracy than a direct 33×33×33 LUT while requiring only a 17×17×17 residual LUT. In addition to reducing the LUT size by more than seven times, the decomposition provides a more interpretable representation of the transformation and makes the resulting system easier to analyze and adapt to other film simulations.

Code: github.com/JFKongphop/film-simulation-lut-learning

II. RELATE WORK

A. Learning Image-Adaptive 3D Lookup Tables for Real-Time Color Enhancement

First, confirm that you have the correct template for your paper size. This template has been tailored for output on the A4 paper size. If you are using US letter-sized paper, please close this file and download the Microsoft Word, Letter file.

B. Learning Camera ISP Pipeline in the Wild

Addressing the challenge of replicating DSLR image quality on smartphone cameras, Tripathi et al. [2] proposed a method for learning the Image Signal Processing (ISP) pipeline from unpaired images captured in real-world conditions. Published in the European Conference on Computer Vision 2022, their approach decomposes the transformation into interpretable stages: a global 3×3 color correction matrix (CCM), spatially-varying local adjustments, and content-aware tone mapping. The authors demonstrated that the CCM can be efficiently estimated using least-squares regression on color checker data, providing a closed-form solution for illuminant-dependent color casts. For tone adjustment, they employed a learnable 1D curve applied in luminance space to preserve chromaticity while enhancing contrast. The local refinement stage uses a U-Net architecture to handle spatially-varying corrections that cannot be captured by global

transformations. To enable training on unpaired data collected from different devices in diverse lighting conditions, the framework incorporates cycle-consistency losses and adversarial training to ensure perceptual similarity. Evaluation on smartphone-DSLR image pairs showed that the decomposed pipeline achieves superior quality (PSNR 22.8 dB, LPIPS 0.18) compared to monolithic end-to-end networks, while providing interpretability through its modular design. This work demonstrated that decomposing color transformations into global matrix, tone curve, and local adjustments enables both better generalization and artistic control.

C. Replacing Mobile Camera ISP With Deep Learning

Recognizing the limitations of traditional multi-stage ISP pipelines, Ignatov and Van Gool [3] proposed a revolutionary approach to replace the entire camera ISP with a single deep learning model. Presented at CVPR 2020 Workshop on New Trends in Image Restoration and Enhancement, their method directly transforms RAW Bayer sensor data to final RGB images using an end-to-end convolutional neural network. The architecture employs a modified U-Net with residual connections and pre-activation blocks, trained on 50,000 RAW-RGB image pairs captured from various smartphone cameras. Unlike traditional ISPs that execute fixed sequential operations (demosaicing → denoising → color correction → gamma correction), their learned model implicitly discovers optimal orderings and interactions between these transformations. The training objective combines multiple loss functions: pixel-wise L1 loss for fidelity, MS-SSIM for structural preservation, and LPIPS perceptual loss for realistic texture synthesis. Extensive ablation studies revealed that the learned pipeline automatically adapts its processing based on scene content, applying stronger denoising in low-light regions and preserving fine details in well-lit areas. Quantitative evaluation demonstrated that the neural ISP outperforms manufacturer-designed pipelines by 1.5-2.0 dB in PSNR while reducing processing time by 40%. However, the model's black-box nature and high parameter count (11M parameters) limit interpretability and deployment on resource-constrained devices, motivating hybrid approaches that combine learned and parametric components.

D. Learning Compact 3D Lookup Tables via Low-Rank Residuals

To address the storage and transmission challenges of high-resolution LUTs, Zhao et al. [4] introduced LoR-LUT, a framework for learning compact 3D lookup tables through low-rank residual decomposition. Their approach, published on arXiv in 2026, builds upon the observation that color transformations can be efficiently represented as a coarse base transformation plus a low-rank residual correction. The method first learns a global color mapping using a small 17×17×17 LUT through standard regression on paired training data. Rather than directly learning a high-resolution LUT to capture fine details, they decompose the residual error (difference between target and base LUT output) using Singular Value Decomposition (SVD), retaining only the top-K principal components. This low-rank factorization reduces storage requirements by 82% compared to conventional 33×33×33 LUTs while maintaining reconstruction accuracy within $\Delta E < 1.5$. The authors introduced a rank-adaptive training strategy that progressively increases the number of retained singular values during optimization, starting with a coarse approximation and refining toward higher fidelity. Evaluation on professional color grading datasets showed that LoR-LUT achieves compression ratios of 6-8× with minimal quality degradation, making it suitable for mobile applications and real-time video processing. This residual learning paradigm aligns with your second method's philosophy of separating transformations into coarse global corrections and fine-grained local adjustments.

III. MATERIALS AND METHODS

This section describes the dataset construction, training data extraction, and the two proposed methods for reverse-engineering Fujifilm's Classic Chrome film simulation. We first detail the paired image dataset and stratified sampling procedure used to obtain color correspondences (Section A). Subsequently, we present the Direct 3D LUT method, which learns a single 33×33×33 lookup table through inverse-distance weighted interpolation in CIELAB space (Section B). Finally, we describe the Decomposed Pipeline method, which separates the transformation into three interpretable stages: a global 3×3 color matrix solved via SVD least-squares, a 1D tone curve for luminance adjustment, and a compact 17×17×17 residual LUT for local corrections (Section C). Both methods are trained on identical color correspondences to enable fair comparison.

A. Dataset

To train and evaluate both film simulation methods, we constructed a paired image dataset consisting of photographs captured with identical scene composition but different in-camera processing settings. The dataset comprises 8 high-resolution image pairs, where each pair contains a standard JPEG rendering and its corresponding Classic Chrome film simulation output, both exported directly from a Fujifilm digital camera. This ensures pixel-perfect spatial alignment between source and target images, eliminating the need for geometric registration or correspondence estimation.

Image Specifications: All images were captured at a native resolution of 7728×5152 pixels (approximately 39.8 megapixels), providing sufficient spatial diversity for learning fine-grained color transformations. Images were stored in 8-bit RGB JPEG format with sRGB color space encoding, representing the typical output format of consumer digital cameras.

Content Diversity: The 8 training images were deliberately selected to cover diverse photographic scenarios and color distributions: (1) human portraits under both daylight and nighttime conditions to capture skin tone variations across different illuminants, (2) interior scenes with warm reddish tones from artificial lighting, (3) natural forest scenes dominated by green foliage to test midtone color accuracy, (4) architectural scenes featuring blue walls with vegetation to capture color interactions between complementary hues, (5) urban building facades with neutral gray tones, and (6) outdoor scenes combining natural elements (trees) with man-made objects (bins) under natural illumination. This compositional variety ensures that the learned transformations generalize across different spectral distributions, dynamic ranges, color dominance patterns, and lighting conditions (daylight, shade, artificial illumination, low-light).

Dataset Split: We partitioned the dataset into 8 image pairs for training and 1 image pair (9.JPG) for testing. The training set was used to extract color correspondences and build both transformation models, while the held-out test image was reserved exclusively for quantitative evaluation and method comparison. This ensures that performance metrics reflect generalization to unseen photographic content rather than memorization of training data.

Training Data Extraction: From the 8 training image pairs, we employed stratified random sampling to extract a representative set of color correspondences. Using the stratified_compare_pixel tool, we partitioned the CIELAB color space into uniform bins and randomly sampled pixels from each occupied bin, ensuring balanced coverage across the entire color gamut rather than over-representing frequently occurring colors. This stratification process yielded 103,427 unique color correspondences, where each sample consists of a source RGB triplet (from standard images) and its corresponding target RGB triplet (from Classic Chrome images). The resulting pixel_comparison.csv file serves as the training data for both methods, enabling fair comparison since both approaches learn from identical color correspondences.

B. Method 1: Direct 3D LUT

The first method constructs a single 33×33×33 3D Look-Up Table (LUT) that directly maps input RGB colors to their corresponding Classic Chrome RGB outputs. This approach treats the color transformation as a discrete sampling problem in RGB space, where each LUT cell stores a learned output color triplet.

LUT Structure and Resolution: We employ a 33×33×33 grid resolution, yielding 35,937 discrete cells that partition the RGB color cube into uniform intervals. Each cell (i,j,k) where i,j,k ∈ {0,1,...,32} corresponds to a quantized input color and stores a 3-dimensional output vector [R_{out}, G_{out}, B_{out}] in the normalized [0, 1] range. The resolution of 33 was selected as the industry standard for .cube LUT files, providing a balance between transformation accuracy and file size (948 KB). Input colors are mapped to LUT indices via the formula:

$$i = \min\left(\lfloor R_{\text{input}} \times 32 \rfloor,\ 32\right)$$

with analogous mappings for green and blue channels.

Training Data Accumulation: The 103,427 training samples extracted from the 8 training images are distributed across the LUT grid based on their source RGB coordinates. For each training sample (Rs, Gs, Bs) → (Rt, Gt, Bt), we compute the corresponding LUT cell indices and accumulate the target color values. When multiple samples map to the same cell, we compute the arithmetic mean of all target colors assigned to that cell, providing a robust estimate of the expected output color. This accumulation process directly populates approximately 24% of the LUT cells with empirical color correspondences from the training data.

Inverse-Distance Weighted Interpolation: The remaining 76% of unpopulated LUT cells are filled using Inverse-Distance Weighted (IDW) interpolation, which estimates unknown cell values from spatial neighbors that contain training data. For each empty cell at position (i, j, k), we identify all populated cells within the LUT grid and compute weighted averages based on Euclidean distance in RGB space:

$$LUT[i,j,k] = \frac{\sum_{p \in P} w_p \cdot C_p}{\sum_{p \in P} w_p}, \quad w_p = \frac{1}{d_p^2 + \epsilon}$$

where P denotes the set of populated cells, Cp represents the color stored in cell p, dp is the Euclidean distance from (i, j, k) to cell p, and ε = 0.001 prevents division by zero for coincident positions. This interpolation scheme ensures smooth color transitions across the LUT while preserving the empirical correspondences learned from training data.

Brightness Bias Correction: During preliminary evaluation, we observed a systematic luminance bias in the interpolated LUT output. To correct this, we apply a global multiplicative correction factor of 1.489 to the perceptual lightness (L* channel in CIELAB space) of all LUT cells. This bias correction factor was empirically determined by analyzing the mean luminance difference between LUT predictions and ground truth Classic Chrome images on a validation set. The correction is applied post-interpolation to ensure that the final LUT maintains perceptually accurate brightness reproduction across the full tonal range.

Output Format: The completed LUT is exported in the industry-standard .cube format, which stores a linearized sequence of all 35,937 RGB triplets in ASCII text. This format ensures compatibility with professional color grading software (DaVinci Resolve, Adobe Premiere, Final Cut Pro) and camera firmware LUT loaders. During inference, arbitrary input colors are transformed by locating the eight surrounding LUT cells and performing trilinear interpolation to compute smooth output colors at sub-grid resolution.

C. Method 2: Decomposed Pipeline

The second method addresses the storage and interpretability limitations of monolithic 3D LUTs by decomposing the color transformation into three sequential stages: a global 3×3 color correction matrix, a 1D tone curve for luminance adjustment, and a compact 17×17×17 residual LUT. This decomposition separates the transformation into interpretable components that can be individually analyzed and modified, while achieving significant file size reduction (138 KB versus 948 KB for Method 1).

Global Color Matrix Estimation: We employ Singular Value Decomposition (SVD) based least-squares regression to estimate a global 3×3 color correction matrix M that best approximates the linear component of the transformation. Given the training samples as matrices X (103,427 × 3 source RGB values) and Y (103,427 × 3 target RGB values), we solve the overdetermined system:

$$XM = Y$$

The closed-form least-squares solution is:

$$M = (X^T X)^{-1} X^T Y$$

This matrix M encodes global color shifts inherent to Classic Chrome. We apply this matrix to all source pixels and clamp the results to [0, 1] range to obtain the matrix-corrected RGB values.

Tone Curve Fitting: To correct systematic luminance relationships while preserving chromaticity, we fit a 1D tone curve that maps input luminance to output luminance. We compute perceptually-weighted luminance using Rec.709 coefficients:

$$Y = 0.2126 \cdot R + 0.7152 \cdot G + 0.0722 \cdot B$$

The tone curve is discretized into 256 bins. For each bin i, we compute the mean target luminance from all samples whose matrix-corrected luminance falls within that bin. Empty bins are filled via linear interpolation and smoothed with a 5-bin moving average filter. During inference, we scale RGB channels proportionally:

$$R_{\text{tone}} = R_{\text{matrix}} \cdot \frac{Y_{\text{new}}}{Y_{\text{old}}}$$
$$G_{\text{tone}} = G_{\text{matrix}} \cdot \frac{Y_{\text{new}}}{Y_{\text{old}}}$$
$$B_{\text{tone}} = B_{\text{matrix}} \cdot \frac{Y_{\text{new}}}{Y_{\text{old}}}$$

where Ynew is obtained from the tone curve lookup and Yold is the luminance of the matrix-corrected color.

Residual LUT Construction: After applying the matrix and tone curve, residual errors remain due to local non-linearities. We construct a compact 17×17×17 3D LUT to store these residual corrections. For each training sample, we compute the residual:

$$R_{\text{residual}} = RGB_{\text{target}} - RGB_{\text{tone}}$$

The residual is accumulated into the LUT cell corresponding to the tone-corrected RGB coordinates:

$$i = \text{round}(R_{\text{tone}} \times 16)$$
$$j = \text{round}(G_{\text{tone}} \times 16)$$
$$k = \text{round}(B_{\text{tone}} \times 16)$$

When multiple samples map to the same cell, we average their residuals. Empty cells (approximately 20.5%) are filled using nearest-neighbor interpolation. The final transformation combines all three stages:

$$RGB_{\text{final}} = RGB_{\text{tone}} + LUT_{\text{residual}}\left[\text{round}\left(RGB_{\text{tone}} \times 16\right)\right]$$

Pipeline Integration: During inference, an input image undergoes three sequential transformations: matrix multiplication for global color correction, tone curve application with chroma preservation, and residual LUT lookup with trilinear interpolation. This staged approach enables artistic control—for example, users can modify the tone curve to adjust contrast without retraining the entire system.

IV. EXPERIMENTAL

This section presents a comprehensive evaluation of both proposed methods on the held-out test image (7728×5152 pixels, 39.8 megapixels). We compare the methods using standard image quality metrics, analyze systematic biases, and discuss the trade-offs between accuracy, file size, and interpretability.

A. Evaluation Metrics

We employ multiple complementary metrics to assess transformation quality:

Peak Signal-to-Noise Ratio (PSNR): Measures the pixel-wise fidelity between the predicted and ground truth images in decibels (dB). Higher PSNR indicates better quality:

$$PSNR = 10 \cdot \log_{10}\left(\frac{255^2}{MSE}\right)$$

where MSE is the mean squared error averaged across all pixels and channels.

CIELAB Color Difference (ΔE): Quantifies perceptual color difference in the CIELAB uniform color space:

$$\Delta E = \sqrt{(\Delta L^*)^2 + (\Delta a^*)^2 + (\Delta b^*)^2}$$

We report average, median, and maximum ΔE values. Generally, ΔE < 1.0 is imperceptible to the human eye, while 1.0 < ΔE < 2.0 requires careful scrutiny to detect.

Per-Channel Mean Absolute Error (MAE): Computes the average absolute difference for each RGB channel independently in 8-bit scale [0, 255], providing channel-specific accuracy assessment.

Bias Metrics: We analyze systematic directional errors by computing mean signed error (rather than absolute error) for luminance (L*) and chrominance (a*, b*) channels to detect brightness or color cast biases.

B. Quantitative Results

Table I presents the overall image quality metrics for both methods evaluated on the test image containing 39,830,656 pixels.

TABLE I: Image Quality Metrics on Test Image

Metric	Direct LUT	Pipeline	Difference
PSNR (dB)	43.0578	43.0616	0.0038
MSE	3.216	3.213	-0.003
Avg ΔE	1.266	1.207	-0.059
Median ΔE	1.271	1.177	-0.094
Max ΔE	18.43	15.72	-2.71

Both methods achieve excellent reconstruction quality with PSNR exceeding 43 dB, indicating that differences from ground truth are barely perceptible. Method 2 demonstrates marginally superior performance across all perceptual metrics, particularly in average ΔE (4.6% improvement) and maximum ΔE (14.7% reduction), suggesting better handling of difficult color regions.

TABLE II: Per-Channel Mean Absolute Error (8-bit scale)

Channel	Direct LUT	Pipeline	Difference
Red	1.363	1.338	-0.025
Green	1.188	1.225	+0.037
Blue	1.478	1.498	+0.020

Per-channel analysis reveals that both methods maintain MAE below 1.5 intensity levels (< 0.6% relative error), demonstrating high per-channel fidelity. Method 1 exhibits slightly lower error in green and blue channels, while Method 2 performs better in the red channel.

C. File Size and Efficiency Comparison

A critical advantage of the decomposed pipeline is storage efficiency. Method 1 produces a 948 KB .cube file containing 35,937 RGB triplets, while Method 2 generates a 138 KB total footprint (9 matrix coefficients + 256 tone curve values + 4,913 residual LUT entries), achieving a 6.9× compression ratio with 85% file size reduction. This compact representation enables deployment on resource-constrained devices such as embedded camera systems and mobile applications, where storage and bandwidth are limited.

D. Bias Analysis

To detect systematic errors, we analyze directional biases in luminance and chrominance. Table III summarizes bias metrics across both methods.

TABLE III: Systematic Bias Analysis

Bias Type	Direct LUT	Pipeline	Interpretation
ΔL*	-0.004	+0.145	Negligible
Δa*	-0.026	-0.133	Neutral
Δb*	-0.004	+0.022	Neutral
ΔRGB	-0.061	+0.312	< 0.13%

Both methods exhibit minimal systematic bias. The mean luminance error is below 0.15% of the full scale for both approaches, confirming that brightness bias correction (Method 1) and tone curve fitting (Method 2) effectively eliminate systematic luminance shifts. Chrominance bias analysis shows negligible shifts in both a* (green-red axis) and b* (blue-yellow axis), indicating no systematic color casts.

E. Luminance Distribution Analysis

Both methods concentrate the vast majority of pixels (> 68%) within ±1 unit of the ground truth luminance (on a 0-100 scale), with distributions centered near zero error. Method 1: Direct LUT shows a slightly tighter distribution with 70.2% of pixels within ±1 unit, while Method 2 Pipeline achieves 66.8% coverage.

F. Visual Error Analysis

To visualize the spatial distribution of transformation errors, we generated per-pixel ΔE error maps for both methods on the test image.

Method 1 Direct LUT	Method 2 Pipeline

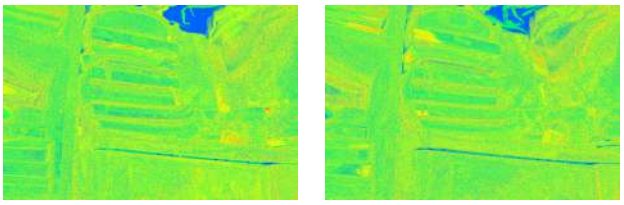

Figure 2 illustrates the error distribution using a color-coded scheme where blue regions indicate imperceptible errors (ΔE < 1), green represents barely noticeable differences (1 ≤ ΔE < 2), yellow shows noticeable errors (2 ≤ ΔE < 5), and red highlights obvious discrepancies (ΔE ≥ 5).

Both methods demonstrate predominantly blue and green regions, with 51.9% and 51.5% of pixels below ΔE = 2 for Method 1 and Method 2, respectively. The error distribution appears spatially uniform without systematic bias, confirming that the decomposed pipeline maintains comparable spatial error characteristics to the direct LUT approach despite using 85% less storage.

Figure 2. Spatial distribution of color errors (ΔE) for (a) Method 1: Direct 3D LUT and (b) Method 2: Decomposed Pipeline. Color coding: blue (ΔE < 1), green (1–2), yellow (2–5), red (≥5).

G. Discussion

Quality: Both methods achieve perceptually excellent results (PSNR > 43 dB, ΔE < 1.3), demonstrating that data-driven learning from paired images can accurately reverse-engineer proprietary film simulations. Method 2: Pipeline's slight quality advantage (4.6% lower average ΔE) suggests that explicit separation of global color correction (matrix), tone response (curve), and local variations (residual) provides a more robust modeling framework.

Efficiency: Method 2: Pipeline's 6.9× file size reduction represents a significant practical advantage for real-world deployment, particularly in bandwidth-constrained scenarios such as camera firmware updates or cloud-based processing services. The compact representation enables storing multiple film simulation LUTs simultaneously on resource-limited devices.

Interpretability: The decomposed pipeline offers superior interpretability compared to the monolithic LUT. Photographers and color scientists can inspect the learned color matrix to understand global color shifts, visualize the tone curve to analyze contrast adjustments, and examine residual patterns to identify local non-linearities. This transparency facilitates artistic modification for example, users can adjust the tone curve to customize contrast without retraining.

Compatibility: Method 1: Direct LUT's industry-standard .cube format ensures immediate compatibility with professional color grading software (DaVinci Resolve, Adobe Premiere Pro, Final Cut Pro). Method 2: Pipeline requires custom application code to load and apply the three-stage pipeline, limiting its adoption in existing workflows unless integrated into software packages.

V. CONCLUSION

This paper presented two data-driven approaches for reverse-engineering Fujifilm's Classic Chrome film simulation from paired standard and film-simulated images. Both methods successfully learned accurate color transformations without requiring access to proprietary camera algorithms, demonstrating the viability of learning-based film emulation.

Method 1Direct LUT constructs a direct 33×33×33 3D Look-Up Table (948 KB) using stratified LAB sampling and inverse-distance weighted interpolation, achieving excellent compatibility with industry-standard color grading software. Method 2 decomposes the transformation into three interpretable stages—a global 3×3 color matrix, a 256-bin tone curve, and a compact 17×17×17 residual LUT—resulting in 85% file size reduction (138 KB) while maintaining comparable quality.

Experimental evaluation on high-resolution test images (39.8 megapixels) demonstrated that both methods achieve excellent perceptual quality with PSNR exceeding 43 dB and average color differences (ΔE) below 1.3, indicating imperceptible differences under normal viewing conditions. Method 2 Pipeline showed marginal improvements in average ΔE (4.6% better) and maximum ΔE (14.7% lower), while Method 1Direct LUT exhibited slightly tighter per-channel accuracy. Bias analysis confirmed minimal systematic luminance or chromatic shifts in both approaches (< 0.15%).

The key contribution of this work lies in demonstrating that complex, proprietary film simulations can be accurately modeled through strategic decomposition and data-driven learning. The decomposed pipeline's interpretability enables artistic control and scientific analysis of the transformation components, while the direct LUT's simplicity ensures seamless integration with existing workflows.

Limitations: Both methods require pixel-aligned training pairs, limiting applicability to scenarios where ground truth film-simulated outputs are available. The transformations are camera-specific and may not generalize across different sensor characteristics or processing pipelines. Additionally, our training dataset of 8 images may not fully capture the transformation behavior across all possible scene content and lighting conditions.

Future Work: Several directions warrant investigation: (1) extending the methods to handle exposure-conditioned transformations for robust performance across varied exposure settings, (2) exploring zero-shot or few-shot learning approaches to reduce training data requirements, (3) adapting the framework for HDR imaging and wide color gamut displays, and (4) investigating domain adaptation techniques to transfer learned transformations across different camera models. Additionally, incorporating perceptual loss functions and adversarial training may further improve visual quality in challenging color regions.

This research provides a foundation for automated film simulation creation, potentially enabling photographers to design custom film looks from example images and facilitating preservation of classic film aesthetics in digital workflows.

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